

Honours Analysis 1 Tutorials (Sec. 003)

Edward Chernysh

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1 Fourth Tutorial

We begin with a rapid review of the definitions. Let X be a subset of $\mathbb{R}$ (a priori, possibly empty). A real number u is said to be an *upper bound* for X if $x \leq u$ for all $x \in X$. If X has an upper bound, we say that X is bounded from above. We call a point $s \in \mathbb{R}$ a *least upper bound* for X if both of the following hold:

- (I) s is an upper bound for X ;
- (II) if u is any upper bound for X , then $s \leq u$.

A given subset of $\mathbb{R}$ may or may not have a least upper bound. If one exists, however, it must be unique. To see this, let s and s' be least upper bounds for a set X . In particular, they are both upper bounds. Thus, we must have $s \leq s'$ and $s' \leq s$ by (II) above. Of course, this forces $s = s'$.

Completeness Property of $\mathbb{R}$. Let X be a non-empty subset of $\mathbb{R}$ that is bounded from above. Then X has a least upper bound, which we denote by $\sup X$.

In your assignment, you have proven that a non-empty subset of $\mathbb{R}$ that is bounded from below has an *infimum*, i.e. a greatest lower bound. If X is such a set, we denote this quantity by $\inf X$. As before, $\inf X$ can be shown to be unique. With this in mind, we can state a converse to the completeness property of $\mathbb{R}$.

Proposition 1. *Let X be a subset of $\mathbb{R}$.*

1. *If X has a least upper bound (i.e. a supremum), then X is non-empty and bounded from above.*
2. *If X has a greatest lower bound (i.e. an infimum), then X is non-empty and bounded from below.*

Proof. We only prove the first claim, and leave the second as an exercise. Suppose that X has a supremum s in $\mathbb{R}$. Then, s is an upper bound for X . By definition, X must be bounded from above. All that remains is to check that X is non-empty. For this, it is enough to show that the empty set has no least upper bound.

Consider the empty set $\emptyset$ and suppose (for a contradiction) that $\emptyset$ has a least upper bound u in $\mathbb{R}$. Then, $x \leq u$ for all $x \in \emptyset$. But, we also have $x \leq u - 1$ for all $x \in \emptyset$. This means that $u - 1 < u$ is an upper bound for $\emptyset$, contradicting the choice of u as the least upper bound. $\square$

Let X be a non-empty set that is bounded from above and let s denote its supremum in $\mathbb{R}$. Let $\varepsilon > 0$ be given and note that $s - \varepsilon < s$ cannot be an upper bound for X (since s is the *least* upper bound for X). Thus, there must exist $x_\varepsilon \in X$ such that $s - \varepsilon < x_\varepsilon$. Since s is an upper bound for X , this gives

$$s - \varepsilon < x_\varepsilon \leq s. \quad (1.1)$$

Conversely, let s be an upper bound for X and suppose that for each $\varepsilon > 0$ one can find $x_\varepsilon \in X$ such that (1.1) holds true. We claim that $s = \sup X$. As s is an upper bound for X , it is enough to show that $s \leq u$ for all upper bounds u of X . If u is an upper bound for X with $u < s$, then $\varepsilon := (s - u) > 0$ whence there exists $x_\varepsilon \in X$ with

$$s - \varepsilon < x_\varepsilon \leq s.$$

But, $s - \varepsilon = s - (s - u) = u$. This gives $u < x_\varepsilon$ which contradicts the fact that u is an upper bound for X . Thus, we must have $s \leq u$. Since u was an arbitrary upper bound, we have $s = \sup X$. This gives the following result from Bartle.

Theorem 1.1. *Let X be a non-empty set and s an upper bound for X in $\mathbb{R}$. The following statements are equivalent.*

1. $s = \sup X$;
2. for every $\varepsilon > 0$ there exists $x_\varepsilon \in X$ such that $s - \varepsilon < x_\varepsilon \leq s$.

For the infimum, a similar argument to what we used above yields an analogous result.

Theorem 1.2. *Let X be a non-empty set and v an lower bound for X in $\mathbb{R}$. The following statements are equivalent.*

1. $v = \inf X$;
2. for every $\varepsilon > 0$ there exists $x_\varepsilon \in X$ such that $v \leq x_\varepsilon < v + \varepsilon$.

It's probably best if we do at least one worked example.

Problem 1. *Consider the set*

$$X = \left\{ \frac{1}{n} - \frac{1}{m} : n, m \in \mathbb{N} \right\}.$$

Compute $\sup X$ and $\inf X$.

Solution. We first check that X is bounded (bounded from above and below). By the triangle inequality, for all $n, m \in \mathbb{N}$:

$$\left| \frac{1}{n} - \frac{1}{m} \right| \leq \frac{1}{n} + \frac{1}{m} \leq 1 + 1 = 2.$$

Hence, X is indeed bounded. By the completeness of $\mathbb{R}$ (and your homework problems), both $\sup X$ and $\inf X$ exist in $\mathbb{R}$. Do note that

$$\frac{1}{n} - \frac{1}{m} \leq \frac{1}{n} \leq 1,$$

for all $n, m \in \mathbb{N}$. Hence, 1 is an upper bound for X . Informally, we should try to look for the largest elements in X and see what they approach. Here, we should take $n = 1$ and try to make $\frac{1}{m}$ as small as possible. In doing so, $1 - \frac{1}{m}$ seems to approach 1. With this in mind, a good idea is to try and show that $\sup X = 1$.

Since we have already shown that 1 is an upper bound, it remains to check that 1 is the least upper bound. Let $\varepsilon > 0$ be given, we want to show that there exists $x_\varepsilon \in X$ such that

$$1 - \varepsilon < x_\varepsilon \leq 1.$$

Choose $m \in \mathbb{N}$ such that $\frac{1}{m} < \varepsilon$ (how?) and note that

$$x_\varepsilon := 1 - \frac{1}{m} \in X.$$

Then, $-\varepsilon < -\frac{1}{m}$ whence

$$1 - \varepsilon < 1 - \frac{1}{m} = x_\varepsilon \leq 1.$$

By Theorem 1.1, we see that $\sup X = 1$. For $\inf X$, we will use a trick from your homework assignment. Observe that

$$X = \left\{ \frac{1}{n} - \frac{1}{m} : n, m \in \mathbb{N} \right\} = \left\{ \frac{1}{m} - \frac{1}{n} : n, m \in \mathbb{N} \right\} = -X.$$

Hence, by Assignment 3 we get $\inf X = \inf (-X) = -\sup X = -1$. $\square$

Given a subset X of $\mathbb{R}$, we define $\max X$ to be the largest element of X , whenever it exists. Similarly, we define $\min X$ as the smallest element of X , whenever it exists. Note that, when these exist, we have $\min X, \max X \in X$.

Problem 2. *Let X be a subset of $\mathbb{R}$. Then $\max X$ exists if and only if $\sup X$ exists and is an element of X . Similarly, $\min X$ exists if and only if $\inf X$ exists and is an element of X .*

Proof. We will only prove the first statement, leaving the second as a potential exercise. Suppose that $\max X$ exists and denote it by m . By definition of $\max X$, we have $m \in X$. Especially, $X \neq \emptyset$. Then, $x \leq m$ for all $x \in X$ whence X is bounded from above. Therefore, $\sup X$ exists. Since $m \in X$, we get $m \leq \sup X$. On the other hand, m is an upper bound for X . By definition of $\sup X$ as the least upper bound, we also have $\sup X \leq m$. It follows that $\sup X = m$ whence $\sup X \in X$.

Conversely, suppose that $\sup X$ exists and is an element of X . By definition, $x \leq \sup X$ for all $x \in X$. Since $\sup X \in X$, this means that $\sup X$ is the maximal element of X . Thus, $\max X$ exists. $\square$

Remark 1.1. This proof tells us that $\max X = \sup X$, whenever $\max X$ exists. Similarly, $\inf X = \min X$ when the latter exists.

1.1 The Existence of $\sqrt{2}$

We will elaborate on the argument used in Theorem 2.4.7 of Bartle's *Introduction to Real Analysis*. We claim that there exists a real number x such that $x^2 = 2$. To show this, we consider the set

$$X = \{s \in [0, \infty) : s^2 < 2\}.$$

Clearly, X is non-empty for $1^2 < 2$ implies $1 \in X$. Also, if $s > 2$ then $s^2 > 4$ implies $s \notin X$. Hence, $s \leq 2$ for all $s \in X$. This means that X is a non-empty bounded subset of $\mathbb{R}$. It therefore has a supremum x in $\mathbb{R}$. Since $1 \in X$ and x is an upper bound for X , we immediately have

$$\boxed{0 < 1 \leq x.}$$

We will show that both cases $x^2 < 2$ and $x^2 > 2$ are impossible. Clearly, this will force $x^2 = 2$ whence x is a square-root of 2.

First, suppose that $x^2 < 2$. For every $n \in \mathbb{N}$ we have $\frac{1}{n^2} \leq \frac{1}{n}$ so that

$$\left(x + \frac{1}{n}\right)^2 = x^2 + \frac{2x}{n} + \frac{1}{n^2} \leq x^2 + \frac{2x+1}{n}. \quad (1.2)$$

By the assumption that $x^2 < 2$, we see that

$$\frac{2-x^2}{2x+1} > 0.$$

Thus, we can choose $n \in \mathbb{N}$ such that

$$0 < \frac{1}{n} < \frac{2-x^2}{2x+1}.$$

This implies that

$$\frac{2x+1}{n} < 2-x^2.$$

Returning to (1.2) yields

$$\left(x + \frac{1}{n}\right)^2 = x^2 + \frac{2x}{n} + \frac{1}{n^2} \leq x^2 + \frac{2x+1}{n} < x^2 + 2 - x^2 = 2. \quad (1.3)$$

Because $x > 0$, we get that $x + \frac{1}{n}$ is an element of X . Since x is an upper bound for X we must have $x + \frac{1}{n} \leq x$, which is a contradiction. Therefore, the case $x^2 < 2$ is impossible.

Next, we show that the case $x^2 > 2$ is impossible. Since we know that $x \geq 1$ and $x^2 > 2$, we actually have $x > 1$. Given $m \in \mathbb{N}$ we calculate

$$\left(x - \frac{1}{m}\right)^2 = x^2 - \frac{2x}{m} + \frac{1}{m^2} > x^2 - \frac{2x}{m}. \quad (1.4)$$

Since $x^2 > 2$, $\frac{x^2-2}{2x}$ is positive. We can therefore choose $m \in \mathbb{N}$ such that

$$0 < \frac{1}{m} < \frac{x^2-2}{2x} \iff 0 < \frac{2x}{m} < x^2-2.$$

Of course, this implies

$$2 - x^2 < -\frac{2x}{m}.$$

Using this choice of m in (1.4), we obtain

$$\left(x - \frac{1}{m}\right)^2 > x^2 - \frac{2x}{m} > x^2 + 2 - x^2 = 2.$$

Since $x > 1$, we have $x - \frac{1}{m} > 0$. If $s > x - \frac{1}{m}$ then

$$s^2 > \left(x - \frac{1}{m}\right)^2 > 2.$$

Hence, $s \leq x - \frac{1}{m}$ for all $s \in X$. This means that $x - \frac{1}{m}$ is an upper bound for X , contradicting the least upper bound property of $\mathbb{R}$.

1.2 The Density of the Irrationals

Recall that a non-empty set $A \subseteq \mathbb{R}$ is said to be dense in $\mathbb{R}$ if, for each $x < y$ in $\mathbb{R}$, there exists $a \in A$ with

$$x < a < y.$$

You have seen in class that the set of rational numbers $\mathbb{Q}$ is dense in $\mathbb{R}$. Here, we show that the irrationals are also dense in $\mathbb{R}$. First, let us recall the proof that $\mathbb{R}$ is uncountable. Typically, one applies the diagonal argument to show that the closed interval $[0, 1]$ is uncountable. From there, it follows that $\mathbb{R} \supset [0, 1]$ is also uncountable. Note that $(0, 1)$ cannot be countable, for then $[0, 1] = (0, 1) \cup \{0, 1\}$ would also be. For any $a < b$ in $\mathbb{R}$, the function

$$f : (0, 1) \rightarrow (a, b), \quad x \mapsto (b-a)x + a$$

is a bijection. Therefore, every open interval (a, b) (with $a, b \in \mathbb{R}$) is uncountable.

Proposition 2. *The set of irrational numbers is dense in $\mathbb{R}$.*

Proof. Let $x, y \in \mathbb{R}$ be such that $x < y$. We want to show that there exists an irrational number $\xi \in (x, y)$. Let $\mathbb{Q}_x^y$ denote the set of rational numbers in the interval (x, y) , i.e.

$$\mathbb{Q}_x^y := (x, y) \cap \mathbb{Q}.$$

As a subset of a countable set, $\mathbb{Q}_x^y$ is countable. Since (x, y) is uncountable, the set $(x, y) \setminus \mathbb{Q}_x^y$ cannot be countable. Put otherwise, $(x, y) \setminus \mathbb{Q}_x^y$ must be uncountable and, in particular, non-empty. Let ξ be any element of $(x, y) \setminus \mathbb{Q}_x^y$. Then,

$$x < \xi < y \quad \text{and} \quad \xi \notin (x, y) \cap \mathbb{Q}.$$

Thus, ξ is an irrational number with $x < \xi < y$. □

Theorem 1.3 (Characterization of Intervals). *Let $S \subseteq \mathbb{R}$ contain at least two points and assume that S satisfies the property*

$$\text{if } x, y \in S \quad \text{and} \quad x < y \quad \text{then} \quad [x, y] \subseteq S.$$

Then, S is an interval.

Proof. There are 4 distinct cases to consider.

- (i) S is bounded;
- (ii) S is bounded from above but not from below;
- (iii) S is bounded from below but not from above;
- (iv) S is unbounded from above and below.

We begin with (i). Define $a := \inf S$ and $b := \sup S$ and observe that $S \subseteq [a, b]$. We now claim that $(a, b) \subseteq S$. Let $z \in (a, b)$ be given. Then, z is not a lower bound for S . We can therefore choose a point $x \in S$ with $x < z$. Similarly, z is not an upper bound for S and there must exist $y \in S$ with $z < y$. But then, $z \in [x, y] \subseteq S$. Since z was arbitrary, we obtain $(a, b) \subseteq S$. If $a, b \in S$, then $[a, b] = S$. If $a \in S$ but $b \notin S$, then $[a, b) = S$. Similarly, if $a \notin S$ but $b \in S$ we have $(a, b] = S$. In either case, S is an interval.

For (ii) we will use a similar argument. Define $b := \sup S$ and observe that $S \subseteq (-\infty, b]$. We claim that $(-\infty, b) \subseteq S$. If $z \in (-\infty, b)$ then $z < b$ whence z is not an upper bound for S . Thus, there exists $y \in S$ with $z < y$. On the other hand, z cannot be a lower bound for S (since S is unbounded from below). This means that we can choose $x \in S$ with $x < z < y$. But then, $z \in [x, y] \subseteq S$. It

follows that $(-\infty, b) \subseteq S$. If $b \in S$, then $(-\infty, b] = S$. Otherwise, $(-\infty, b) = S$. In either case, S is an interval.

For (iii) we will “piggy back” off (ii). Suppose that S is bounded from below but not from above. Then, $-S$ is bounded from above but not from below. By (ii), $-S$ must be an interval of the form $(-\infty, b)$ or $(-\infty, b]$. But then, $S = -(-S)$ is of the form

$$(-b, \infty) \quad \text{or} \quad [-b, \infty).$$

Now we handle (iv). Suppose that S is unbounded from above and below. We claim that $(-\infty, \infty) \subseteq S$. Let $z \in (-\infty, \infty)$ be given and note that z is neither a lower bound nor an upper bound for S . Thus, we can find $x, y \in S$ such that $x < z < y$. That is, $z \in [x, y] \subseteq S$. This yields $\mathbb{R} = (-\infty, \infty) \subseteq S \subseteq \mathbb{R}$. Since $S = \mathbb{R}$, we see that S is an interval. $\square$

1.3 Algebraic Properties of the Supremum/Infimum

Problem 3. Let A and B be non-empty subsets of $\mathbb{R}$ that are bounded above. Suppose both X and Y only contain non-negative elements. Show that

$$\sup(A \cdot B) = \sup A \cdot \sup B.$$

Here, $A \cdot B = \{ab : a \in A, b \in B\}$.

Proof. Clearly, $A \cdot B \neq \emptyset$. Since $0 \leq a \leq \sup A$ and $0 \leq b \leq \sup B$, we have

$$0 \leq ab \leq \sup A \cdot \sup B$$

for all $ab \in A \cdot B$. Hence, $A \cdot B$ is bounded from above. This means that $A \cdot B$ has a least upper bound. The above tells us that $A \cdot B$ is bounded above by $\sup A \cdot \sup B$; it only remains to check that this is the least upper bound.

First, we handle the case where $\sup A = 0$ or $\sup B = 0$. Since $0 \leq a \leq \sup A$ and $0 \leq b \leq \sup B$ for all $a \in A$ and $b \in B$, we would have either $A = \{0\}$ or $B = \{0\}$. In either case, $A \cdot B = \{0\}$ whence $\sup(A \cdot B)$ is indeed equal to $0 = \sup A \cdot \sup B$.

Now we check the case where $\sup A$ and $\sup B$ are non-zero (and hence strictly larger than 0). Let u be an upper bound for $A \cdot B$. Then,

$$0 \leq ab \leq u, \quad \forall a \in A, b \in B.$$

Let $b \in B$ with $b \neq 0$. Then,

$$0 \leq a \leq \frac{u}{b}, \quad \forall a \in A.$$

This forces $0 \leq \sup A \leq \frac{u}{b}$ whence $0 \leq b \sup A \leq u$. Since $b \neq 0$ was arbitrary, this actually holds for all $b \in B \setminus \{0\}$. Since it is trivially true for $b = 0$, we have

$$0 \leq b \sup A \leq u, \quad \forall b \in B.$$

But this yields

$$0 \leq b \leq \frac{u}{\sup A}$$

for all $b \in B$ so that $\sup B \leq \frac{u}{\sup A}$. Of course, this gives $\sup A \sup B \leq u$. Since u was any upper bound of $A \cdot B$, we conclude that $\sup(A \cdot B) = \sup A \cdot \sup B$. $\square$

Problem 4. Let X and Y be non-empty subsets of $\mathbb{R}$.

1. If both X and Y are bounded from above, then $\sup(X \cup Y)$ exists and is equal to $\max(\sup X, \sup Y)$.
2. If both X and Y are bounded from below, then $\inf(X \cup Y)$ exists and is equal to $\min(\inf X, \inf Y)$.

Proof. We only prove the second statement, as the first follows from a similar argument. Since $\inf X \leq x$ and $\inf Y \leq y$ for all $x \in X$ and $y \in Y$, it is clear that $\min(\inf X, \inf Y)$ is a lower bound for the non-empty set $X \cup Y$ (both X and Y are non-empty). Hence, $\inf(X \cup Y)$ exists. Let v be a lower bound for $X \cup Y$, we will show that

$$v \leq \min(\inf X, \inf Y).$$

Because $\min(\inf X, \inf Y)$ is a lower bound for $X \cup Y$, this would mean that $\min(\inf X, \inf Y)$ is the greatest lower bound for $X \cup Y$. Since v is a lower bound for $X \cup Y$, we have $v \leq z$ for all $z \in X \cup Y$. In particular, $v \leq x$ and $v \leq y$ for all $x \in X$ and $y \in Y$. This implies that $v \leq \inf X$ and $v \leq \inf Y$. Therefore, $v \leq \min(\inf X, \inf Y)$. $\square$

1.4 More About Functions

Let X be a non-empty set and $f : X \rightarrow \mathbb{R}$ a function. We say that f is bounded from above if $f(X)$ is bounded from above in $\mathbb{R}$. Similarly, f is said to be bounded from below if $f(X)$ is bounded from below in $\mathbb{R}$. If f is bounded from above, we define

$$\sup_{x \in X} f := \sup f(X) = \sup \{y : y = f(x) \text{ for some } x \in X\}.$$

If f is bounded from below, we have the analogous definition:

$$\inf_{x \in X} f := \inf f(X) = \inf \{y : y = f(x) \text{ for some } x \in X\}.$$

Problem 5. Let X and Y be non-empty sets. Let $f : X \rightarrow Y$ be any function and $g : Y \rightarrow \mathbb{R}$ bounded from above. Show that $g \circ f$ is bounded from above and that

$$\sup_{x \in X} (g \circ f) \leq \sup_{y \in Y} g. \quad (1.5)$$

In addition, we have equality when f is surjective.

Proof. The function $g \circ f$ is bounded from above whenever g is bounded from above since $(g \circ f)(X) \subseteq g(Y)$. Therefore,

$$\sup_{x \in X} (g \circ f) = \sup (g \circ f)(X)$$

exists. To prove the desired inequality, it is enough to show that $\sup_{y \in Y} g$ is an upper bound for the set $(g \circ f)(X)$. Clearly, any element of $(g \circ f)(X)$ takes the form $g(f(x))$ for some $x \in X$. Rather, it takes the form $g(y)$ for some element $y = f(x) \in Y$. Thence,

$$(g \circ f)(x) = g(f(x)) = g(y) \leq \sup_{y \in Y} g.$$

We have thus proven (1.5). Now, suppose that f is surjective. This means that $g(Y) = g(f(X))$ whence

$$\sup_{y \in Y} g = \sup g(Y) = \sup g(f(X)) = \sup (g \circ f)(X) = \sup_{x \in X} (g \circ f).$$

See the remark below for an example in which equality fails when f is not surjective. $\square$

Remark 1.2. Equality may fail in (1.5) if f is not surjective. Let $X = Y = [0, 1]$ and define functions $f : X \rightarrow Y, g : Y \rightarrow \mathbb{R}$ according to the formulas

$$f(x) = 0, \quad g(y) = y.$$

Then, $(g \circ f)(X) = \{0\}$ so that

$$\sup_{x \in X} (g \circ f) = 0.$$

On the other hand, $g(Y) = [0, 1]$. By an earlier exercise (Problem 2), we have $\sup g(Y) = \max g(Y) = 1 > 0$. Therefore,

$$\sup_{x \in X} (g \circ f) < \sup_{y \in Y} g. \quad (1.6)$$

Email: edward.chernysh@mail.mcgill.ca

Office: BURN 1122

Office Hours: Wed. 11:00am - 12:00pm & Thurs. 3:30pm - 4:30pm.